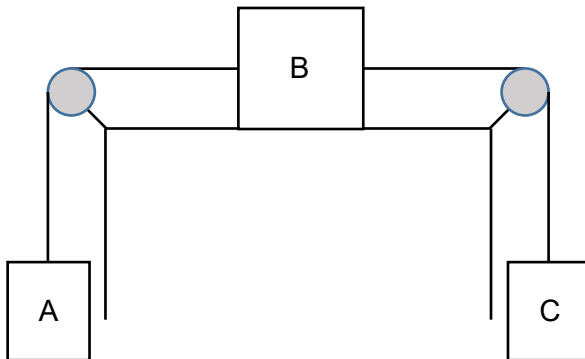


- 5 Three bodies A, B and C with mass 15 kg, 45 kg and 25 kg are connected by light strings that pass over massless and smooth pulleys. The coefficient friction between B and the surface of the table is 0.15.



What is the acceleration of the three bodies?

- A**  $0 \text{ m s}^{-2}$       **B**  $0.38 \text{ m s}^{-2}$       **C**  $1.2 \text{ m s}^{-2}$       **D**  $2.6 \text{ m s}^{-2}$

Let tension in the string connection BC be  $T_1$  and AB be  $T_2$ ,

$$m_C g - T_1 = m_C a$$

$$T_1 - T_2 - \mu m_B g = m_B a$$

$$T_2 - m_A g = m_A a$$

Solving simultaneously,  $a = 0.38 \text{ m s}^{-2}$ .

Ans: B